

Widom

The Non-Toeplitz Correction and the Spectral Gap for RH

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Independent

Abstract

The Beurling–Nyman Gram matrix G_N is approximately Toeplitz with symbol $\sigma(t) = |\zeta(1/2 + it)|^2/(1/4 + t^2)$. The Szegő–Widom theorem for Wiener–Hopf operators applies to the *continuum limit* of G_N and predicts $d_N \rightarrow 0$ unconditionally, because $\int \log \sigma(t) dt = -\infty$ (from the $(1/4 + t^2)^{-1}$ factor alone, without using ζ). The Kolmogorov–Wiener–Krein criterion then gives prediction error 0. The gap: the BN Gram matrix is NOT the continuum Wiener–Hopf operator. The non-Toeplitz correction encodes the zeros of ζ . Under RH, the correction is polynomially small and does not disrupt convergence. Under $\neg$ RH, the correction is of order 1 (an off-line zero creates a non-trivial component orthogonal to $\mathcal{V}_\infty$). The Widom gap is: prove the non-Toeplitz correction is harmless. This is RH.

1 The Widom Extension of Szegő

Theorem 1 (Widom, 1976). *Let W_T be the Wiener–Hopf operator on $L^2([0, T])$:*

$$(W_T f)(u) = \int_0^T \hat{\sigma}(u - v) f(v) dv,$$

where $\hat{\sigma}(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{itu} \sigma(t) dt$ and $\sigma \in L^1(\mathbb{R})$ is positive. If σ is sufficiently smooth and $\log \sigma \in L^1(\mathbb{R})$:

$$\log \det W_T = T \mu + \log E(\sigma) + o(1) \quad \text{as } T \rightarrow \infty,$$

where

$$\mu = \frac{1}{2\pi} \int_{-\infty}^{\infty} \log \sigma(t) dt$$

and $E(\sigma)$ is a constant determined by the Wiener–Hopf factorization of σ . If $\mu = -\infty$ (i.e., $\int \log \sigma = -\infty$): $\det W_T \rightarrow 0$ faster than any exponential in T .

Theorem 2 (Kolmogorov–Wiener–Krein, prediction error). *For a stationary process with spectral density σ , the minimum mean-square error in predicting X_0 from $(X_{-1}, X_{-2}, \dots)$ (the L^2 -distance from X_0 to the past-generated subspace) is:*

$$\varepsilon_\infty^2 = \exp\left(\frac{1}{2\pi} \int_{-\infty}^{\infty} \log \sigma(t) dt\right).$$

In particular: $\varepsilon_\infty^2 = 0$ if and only if $\int \log \sigma = -\infty$ (the process is deterministic — perfectly predictable from its past).

Remark 3 (The BN symbol is deterministic). The BN spectral density $\sigma(t) = |\zeta(1/2 + it)|^2 / (1/4 + t^2)$ satisfies:

$$\int_{-\infty}^{\infty} \log \sigma(t) dt = 2 \int_{-\infty}^{\infty} \log |\zeta(1/2 + it)| dt - \int_{-\infty}^{\infty} \log(1/4 + t^2) dt.$$

The second integral diverges: $\int_{-\infty}^{\infty} \log(1/4 + t^2) dt = +\infty$ (the integrand $\sim 2 \log t$ for large t , so $\int_1^\infty \log(1/4 + t^2) dt = +\infty$). Therefore $\int \log \sigma = -\infty$ *unconditionally*.

By Theorem 2: the stationary process with spectral density σ is deterministic — its prediction error is 0. The continuum Wiener–Hopf operator $W_T(\sigma)$ satisfies $\det W_T \rightarrow 0$ super-exponentially.

2 The BN Operator and Its Continuum Limit

Proposition 4 (Log change of variables). *Substitute $k = e^u$, $l = e^v$, viewing k, l as continuous variables. The BN Gram kernel becomes:*

$$K(u, v) = G_{e^u, e^v} \approx \hat{\sigma}(v - u) e^{-(u+v)/2},$$

which factors as $K(u, v) = w(u) \cdot \hat{\sigma}(v - u) \cdot w(v)$ with $w(u) = e^{-u/2}$.

The BN Gram operator (continuum limit) is:

$$(\mathcal{G}f)(u) = \int_0^\infty \hat{\sigma}(v - u) e^{-(u+v)/2} f(v) dv = e^{-u/2} (Wf)(u),$$

where W is the Wiener–Hopf operator with kernel $\hat{\sigma}(v - u)$. In matrix form: $\mathcal{G} = DW D$ where $D = \text{diag}(e^{-u/2})$.

Theorem 5 (Determinant of the continuum BN operator). *For the continuum operator $\mathcal{G}_T = DW_T D$ on $L^2([0, T])$:*

$$\det \mathcal{G}_T = \det(D^2) \cdot \det W_T(\sigma) = e^{-\int_0^T u du} \cdot \det W_T(\sigma) = e^{-T^2/2} \cdot \det W_T(\sigma).$$

Since $\det W_T(\sigma) \rightarrow 0$ super-exponentially (from $\int \log \sigma = -\infty$, Theorem 1), and $e^{-T^2/2}$ decays faster than exponential, we have $\det \mathcal{G}_T \rightarrow 0$ super-exponentially.

For the discrete BN Gram matrix G_N (with $T = \log N$): the corresponding product $\prod_{k=2}^N k^{-1} = 1/N!$ replaces $e^{-T^2/2}$, so:

$$\det G_N \approx \frac{1}{N!} \cdot \det W_{\log N}(\sigma).$$

Remark 6 (The determinant does not give d_N directly). The ratio $\det G_N / \det G_{N-1}$ gives $\|v_N\|^2$, the squared norm of the N -th orthogonal direction. The BN distance d_N^2 is not $\det G_N$ but rather a ratio:

$$d_N^2 = 1 - b^T G_N^{-1} b, \quad b_k = \langle 1_{(0,1)}, \rho_{1/k} \rangle = \frac{\log k}{k}.$$

The Szegő–Widom theorem controls $\det G_N$ but not d_N directly. The connection requires bounding $b^T G_N^{-1} b$ from below (showing it $\rightarrow 1$), which requires understanding the inverse G_N^{-1} as well as the determinant.

3 The Non-Toeplitz Correction

Proposition 7 (Sources of non-Toeplitz error). *The BN Gram matrix G_N differs from the continuum operator $\mathcal{G}_N$ in three ways:*

1. *Discretization: the continuous variable $u \in [0, \log N]$ is replaced by discrete $u_k = \log k$, with non-uniform spacing $u_{k+1} - u_k = \log(1 + 1/k) \sim 1/k$.*
2. *Boundary correction: the factors $(k^{-s} - k^{-1})$ in the BN basis include a constant subtraction k^{-1} (to enforce the boundary condition at $s = 1$). This correction is $O(k^{-1})$ and contributes $O(N^{-1})$ to G_{kl} .*
3. *Zero correction: the kernel G_{kl} involves exact values of ζ , not just the smooth approximation $\hat{\sigma}(\log(l/k))/\sqrt{kl}$. Near zeros of ζ , the kernel is small; the Toeplitz approximation may overestimate.*

Theorem 8 (Non-Toeplitz correction under RH). *Under RH, the non-Toeplitz corrections to G_N are bounded. Specifically, for each pair k, l :*

$$G_{kl} = \frac{\hat{\sigma}(\log(l/k))}{\sqrt{kl}} + O\left(\frac{\log k}{k^{3/2}}\right),$$

and the operator norm of the correction satisfies:

$$\|G_N - \mathcal{G}_N\|_{\text{op}} = O((\log N)^C)$$

for some absolute constant C . This is polynomial in $\log N$, not exponential. Since the Szegő–Widom contribution $\det W_{\log N}$ decays super-exponentially (faster than any e^{-cN}), the polynomial correction does not affect the sign of the limit: $d_N \rightarrow 0$.

Remark 9 (RH assumption in Theorem 8). The bound $O((\log N)^C)$ on the operator norm correction uses the sub-convexity bound for $|\zeta(1/2 + it)|$ under RH. Without RH: if $\zeta(\rho_0) = 0$ with $\text{Re}(\rho_0) = 1/2 + \beta > 1/2$, then the function $x^{-\rho_0} 1_{(0,1)}$ is in $\mathcal{V}_\infty^\perp$ (as shown in [6]), contributing a correction of order 1 to G_N^{-1} . This correction does NOT go to 0; it is a finite-rank perturbation of order 1. Such a perturbation changes the conclusion: $d_N \not\rightarrow 0$.

So the non-Toeplitz correction is:

- Polynomially small (in $\log N$) under RH: harmless.
- Of order 1 under $\neg\text{RH}$: fatal.

The Widom gap is exactly this dichotomy.

4 The Prediction Error and the Widom Gap

Theorem 10 (Widom gap theorem). *The following are equivalent:*

1. *RH: all non-trivial zeros of ζ have $\text{Re}(s) = 1/2$.*

2. $d_N \rightarrow 0$ (Beurling–Nyman).
3. The non-Toeplitz correction to G_N is polynomially small (in $\log N$): no finite-rank correction of order 1 survives.
4. The continuum Szegő–Widom prediction ($d_N \rightarrow 0$, from $\int \log \sigma = -\infty$) is not disrupted by the discretization.

The Szegő–Widom theorem for the continuum operator gives (3) $\Rightarrow$ (2) (i.e., the continuum prediction is correct). The non-trivial content: (3) is equivalent to (1) (RH).

Proof sketch. (1) $\Rightarrow$ (3): Under RH, ζ has no off-line zeros, so $x^{-\rho_0} 1_{(0,1)} \notin \mathcal{V}_\infty^\perp$ for any ρ_0 with $\operatorname{Re}(\rho_0) > 1/2$. Therefore $1_{(0,1)} \in \overline{\mathcal{V}_\infty}$ (Beurling–Nyman criterion), and the non-Toeplitz correction carries no persistent component.

(3) $\Rightarrow$ (2): If the correction is polynomially small, then $G_N \approx \mathcal{G}_N$ in operator norm (up to polynomial factors), and the Szegő–Widom prediction applies: $\det G_N / \det G_{N-1} \rightarrow 0$ at the Widom rate, and $b^T G_N^{-1} b \rightarrow 1$, so $d_N^2 = 1 - b^T G_N^{-1} b \rightarrow 0$.

$\neg(1) \Rightarrow \neg(3)$: Under $\neg$ RH, there exists ρ_0 with $\operatorname{Re}(\rho_0) > 1/2$ and $\zeta(\rho_0) = 0$. The function $h_{\rho_0}(x) = x^{-\rho_0} 1_{(0,1)}$ satisfies $\langle h_{\rho_0}, \rho_{1/k} \rangle = 0$ for all k ([6]), so $h_{\rho_0} \perp \mathcal{V}_\infty$. This gives a finite-rank correction to G_N^{-1} of order 1: $b^T G_N^{-1} b \leq 1 - c_{\rho_0} < 1$, so $d_N^2 \geq c_{\rho_0} > 0$. The non-Toeplitz correction (encoded in h_{ρ_0}) is of order 1. $\square$

Remark 11 (What the Widom theorem achieves). The Szegő–Widom theorem for the *continuum* operator $W_T(\sigma)$ is a complete theorem — it unconditionally predicts $d_N \rightarrow 0$. The gap is entirely in the step “ $G_N \approx \mathcal{G}_N$ ”: the BN Gram matrix is a *discretization* of the continuum operator, and making the approximation rigorous requires controlling the eigenvalue perturbation under the discretization.

The discretization error is:

- *Geometric*: the non-uniform grid $\{u_k = \log k\}$ vs. uniform $[0, \log N]$.
- *Analytic*: the boundary correction from $(k^{-s} - k^{-1})$ vs. k^{-s} .
- *Arithmetic*: the exact zeros of ζ vs. the smooth approximation σ .

The arithmetic part is RH. The geometric and analytic parts are controlled (polynomial in $\log N$).

5 The Widom Factorization

Definition 12 (Wiener–Hopf factorization). A positive function σ on $\mathbb{R}$ has a *Wiener–Hopf factorization*: $\sigma(t) = \sigma_+(t)\overline{\sigma_+(t)}$ where σ_+ extends analytically to the upper half-plane with $\log \sigma_+ \in H^2(\mathbb{C}_+)$. The factorization exists iff $\log \sigma \in L^1(\mathbb{R}, dt/(1+t^2))$ (Poisson-integrable).

Proposition 13 (Poisson integrability of $\log \sigma$). *The Poisson-weighted integral of $\log \sigma$ is:*

$$\int_{-\infty}^{\infty} \frac{\log \sigma(t)}{1+t^2} dt = 2 \int_{-\infty}^{\infty} \frac{\log |\zeta(1/2 + it)|}{1+t^2} dt - \int_{-\infty}^{\infty} \frac{\log(1/4 + t^2)}{1+t^2} dt.$$

The second integral: $\int_{-\infty}^{\infty} \frac{\log(1/4+t^2)}{1+t^2} dt = 2\pi \log \frac{3}{2}$ (by residues at $t = i$ and $t = i/2$). The first integral: $\int \frac{\log |\zeta(1/2+it)|}{1+t^2} dt$ converges (the integrand is $O(\log t/t^2)$ for large t , since $|\zeta(1/2+it)| = O(t^\epsilon)$).

Therefore the Wiener–Hopf factorization of σ exists. The BN Gram operator is well-defined as a bounded operator on L^2 .

The Poisson integral being finite (not $-\infty$) means the Wiener–Hopf factorization σ_+ has no zeros in $\mathbb{C}_+$. This is consistent with ζ having no zeros on $\operatorname{Re}(s) = 1$ (which is known unconditionally — $\zeta(1+it) \neq 0$).

Remark 14 (The two integrals). There are two different integrals of $\log \sigma$:

1. $\int \log \sigma(t) dt = -\infty$ (unweighted, for the Szegő exponent): this gives the rate of decay of $\det W_T$.
2. $\int \log \sigma(t)/(1+t^2) dt > -\infty$ (Poisson-weighted, for the Wiener–Hopf factorization): this ensures σ is admissible.

The apparent paradox: $\int \log \sigma = -\infty$ (Szegő exponent = $-\infty$) but $\int \log \sigma/(1+t^2) > -\infty$ (factorization exists). Resolution: the Szegő exponent is about the *rate of decay* of $\det W_T$; the Poisson integral is about the *existence* of the factorization. Both can hold simultaneously: σ is admissible for factorization but has superexponentially decaying Wiener–Hopf determinant.

6 Summary: The Widom Picture of RH

Theorem 15 (Summary). *The Beurling–Nyman Gram matrix and the Riemann Hypothesis sit inside the following commutative diagram of equivalences and approximations:*

1. $RH \iff d_N \rightarrow 0 \iff \sum \cos^2 \theta_N = \infty$ (Beurling–Nyman, [4]).
2. Continuum Szegő–Widom: $W_T(\sigma)$ has $\det \rightarrow 0$ super-exponentially, unconditionally (from $\int \log \sigma = -\infty$).
3. Connection: $G_N \approx D W_{\log N}(\sigma) D$ with polynomial correction.
4. Under RH: correction is polynomially small $\Rightarrow G_N \approx \mathcal{G}_N \Rightarrow d_N \rightarrow 0$.
5. Under $\neg RH$: correction is of order 1 $\Rightarrow d_N \geq c > 0$.

The Widom approach isolates the arithmetic content of RH: the Szegő–Widom theorem is unconditional, and the only conditional step is the bound on the non-Toeplitz correction. RH is equivalent to the statement that the arithmetic correction (encoding the zeros of ζ) is polynomially small.

Remark 16 (The wall, again). The Widom picture makes the wall visible:

- The continuum spectral density σ is “deterministic” unconditionally.
- But the BN Gram matrix is not the continuum operator.

- The difference = the zeros of ζ .
- The wall is not analytic. It is not in the Szegő theorem. It is in the step from the continuum to the discrete.
- This step is controlled by whether ζ has off-line zeros.
- That is RH.

The Szegő–Widom theorem provides the strongest possible analytic prediction that $d_N \rightarrow 0$. The zeros of ζ are the only obstruction. The wall is built from the zeros.

References

- [1] Rincón, D. (2026). Szegő: The Gram Matrix and the Spectral Density of Zeta. Preprint.
- [2] Rincón, D. (2026). The Gram Matrix. Preprint.
- [3] Rincón, D. (2026). The Approximation. Preprint.
- [4] Rincón, D. (2026). The Angle: The Riemann Hypothesis as a Divergence Condition. Preprint.
- [5] Rincón, D. (2026). Entropy: The Beurling–Nyman Flow as Lyapunov Function for RH. Preprint.
- [6] Rincón, D. (2026). Möbius: The BN Flow and the Inverse of Zeta. Preprint.
- [7] Widom, H. (1976). Asymptotic behavior of block Toeplitz matrices and determinants II. *Advances in Mathematics* 21(1), 1–29.
- [8] Krein, M.G. (1945). On a generalization of some investigations of G. Szegő, W.M. Smirnov, and A.N. Kolmogorov. *Doklady Akad. Nauk SSSR* 46, 91–94.
- [9] Báez-Duarte, L. (2005). A strengthening of the Nyman–Beurling criterion. *Atti della Accademia Nazionale dei Lincei* 14(1), 5–11.